

CS407 In Class Activity

kNN Regression Bridge

Your group will use the same cleaned dataset from the previous activity to examine how feature scaling, the number of neighbors, and neighbor weighting affect k-nearest neighbors regression. You will compare kNN with ordinary linear regression using the same evaluation procedure used by every group.

Synthetic instructional data All records in this activity were generated for classroom use and do not represent actual students. Associations do not establish that changes in study, sleep, attendance, or another predictor cause changes in scores. Treat the results as model evidence within this synthetic sample.

Format and submission Use your group's existing dataset and the same cleaned predictors and categorical encoding from the previous activity. Do not create a new dataset or a separately encoded data file. Plan to complete most of the work during class. The Group POC must submit one completed notebook before the next class meeting. No formal presentation is required.

Learning goals

- Explain why feature scaling matters for distance-based models.
- Describe how the number of neighbors changes model flexibility.
- Compare kNN with ordinary linear regression using cross-validated metrics.
- Interpret a diagnostic plot and explain why kNN does not produce regression coefficients.

Dataset assignments

Group	Team	Dataset	Variation
1	PEM	Group1_Baseline.csv	Clean baseline
2	Team Terminal	Group2_Sleep_Outlier.csv	One unusual Sleep_Hours observation
3	OMAC	Group3_Music_Genre.csv	Additional categorical feature
4	SKA	Group4_Distance.csv	Additional numeric feature
5	Black Eagle	Group5_Department_Shift.csv	Department-based predictor distributions

Tasks

- 1. Define the modeling problem** Use Exam_Score as the target y . Use the same cleaned predictors X from the previous activity, and exclude Synthetic_Record_ID from every model.
- 2. Predict the effect of scale** Identify the predictor with the largest numerical range. Predict which predictor may dominate Euclidean distance before scaling and whether standardization will change the nearest neighbors.
- 3. Prepare the predictors** Keep the previous categorical setup: OneHotEncoder with drop='first' and handle_unknown='ignore'. Standardize numeric predictors only for scaled kNN, inside the pipeline and within each training fold.
- 4. Run the scaling comparison** Compare ordinary linear regression, kNN with $k=5$ using unscaled numeric predictors, and kNN with $k=5$ using standardized numeric predictors. Record mean and standard deviation for MAE, RMSE, and R^2 .
- 5. Choose the number of neighbors** Evaluate scaled kNN with k values of 1, 3, 5, 9, and 15 using uniform weights. Select the value with the lowest mean RMSE, then use MAE, R^2 , and variability to qualify the decision.
- 6. Compare neighbor weighting** Using the selected k , compare uniform weighting with distance weighting. Explain whether giving closer neighbors more influence improved estimated generalization.
- 7. Diagnose and interpret** Create a predicted-versus-actual or residual plot from cross-validated predictions. Compare the selected kNN model with ordinary linear regression, explain how the dataset variation affected the results, and state one limitation and one non-causal interpretation.

Models to compare

Retain the models below because each answers a different question. The first three isolate the effect of scaling. The selected kNN model then shows whether adjusting k and neighbor weights improves the comparison with the linear baseline.

Model	Preprocessing	Purpose
Linear regression	Existing one-hot encoding; numeric unscaled	Global linear baseline
kNN, $k=5$, unscaled	Existing one-hot encoding; numeric unscaled	Demonstrate scale sensitivity
kNN, $k=5$, scaled	Existing one-hot encoding; numeric standardized	Fair kNN starting point
Selected kNN	Same scaled pipeline with selected k and weights	Final kNN comparison

Common evaluation procedure

Every group must use the settings below. Keeping the procedure fixed allows the class to attribute differences to the datasets and models rather than to different random splits.

Item	Required setting
Outer evaluation	RepeatedKFold with 5 splits, 5 repeats, and random_state=407
Candidate k values	1, 3, 5, 9, and 15 using uniform weights
Weight comparison	Uniform and distance weights using the selected k
Metrics	Mean and standard deviation of MAE, RMSE, and R-squared across the 25 folds
Selection rule	Lowest mean RMSE, with MAE, R-squared, and variability reported as supporting evidence
Prediction plot	Separate shuffled 5-fold KFold with random_state=407 and cross_val_predict
Leakage rule	Fit encoding and scaling inside each pipeline, never on the full dataset before validation

Submission requirements

- One completed notebook submitted by the Group POC before the next class meeting
- A results table comparing linear regression with unscaled and scaled kNN
- A results table for the candidate k values and the two weighting methods
- One predicted-versus-actual or residual plot based on cross-validated predictions
- The selected k and weighting method, with a short model recommendation
- An explanation of how the group-specific variation affected the results
- One limitation, one non-causal interpretation statement, and responses to the final reflection questions

Group specific questions

Group	Question to answer
1	How does scaled kNN compare with the linear baseline, and how does performance change as k increases?
2	How does the unusual Sleep_Hours value affect neighbor distances, its residual, and validation variability?
3	How does one-hot encoding Favorite_Music_Genre affect neighborhoods and validation performance?
4	Does Distance_From_Home_Miles dominate before scaling, and what changes after standardization?
5	How does one-hot encoding Department affect neighborhoods and validation performance?

Lightning debrief Be prepared to give a two minute oral summary of the effect of scaling, the selected k, and the better-performing model. No slides are required.